MIL-T-49247(CR)

24 April 1981

MILITARY SPECIFICATION

TRANSCEIVER MULTIPLEXERS TD-1288()/GRC AND TD-1289()(V)/GRC

This specification is approved for use by the Communications Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specifications covers the following configurations of the 2 Channel Multiplexer TD-1288()/GRC and the 5 Channel Multiplexer TD-1289()(V)/GRC. When not otherwise specified, the term "multiplexer" refers to all multiplexer configurations:

Multiplexer	Coupler CU-2266()/GRC	Coupler CU-2267()/GRC	Bandpass Filter F-1482()/GRC	Termination Unit MX-10080()/GRC
TD-1288()/GRC	1	0, N/A	2	0, N/A
TD-1289()(V)1/GRC	0, N/A	1	5	0, N/A
TD-1289()(V)2/GRC	0, N/A	1	4	1
TD-1289()(V)3/GRC	0, N/A	1	3	2

In addition, this specification covers the transit cases used in transporting the above equipments; i.e., Case, CY-7775()/GRC and Case, CY-7776()/GRC.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in approving this document should be addressed to Commander, US Army Communications Research and Development Command, ATTN: DRDCO-PE-EC-1, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement.

FSC 5805

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2.0 APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of Request for Proposal, term a part of this specification to the extent specified herein. In the event of conflict between requirements expressed in applicable documents and portions of this specification, this specification shall take precedance.

SPECIFICATIONS

Military

MIL-P-116	Preservation - Packaging, Methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-F-14072	Finish for Ground Electronic Equipment
MIL-M-13231	Marking at Electronic Items
MIL-E-55585	Electronic Equipment & Parts, Packaging of
* Appendix A to EL-CP-0192-0001A	Nuclear Survivability (U)

STANDARDS

Federal

Federal Std No. 595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attribution
MIL-STD-252	Wired Equipment Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-726	Packaging Requirements Code
MIL-STD-781C	Reliability Design Qualification and Production Acceptance Test: Exponential Distribution
MIL-STD-810	Environmental Test, Methods

DRAWINGS

Communications Research and Development Command

DL-SM-B-946800	Multiplexer Group 2 Channel
DL-SM-B-946801	Multiplexer Group 5 Channel

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions, should be obtained from the procurement activity or as directed by the Contracting Officer. Both title and number should be stipulated when requesting copies).

*Classified Document

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.1.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and of Drawings and Data Lists DL-SM-B-946800 and DL-SM-B-946801 including the requirements for parts materials and processes thereon.

3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein, shall apply except as noted on Drawing and Data Lists referenced in 3.2. (See 4.4)

3.4 Finish. The equipment shall be finished in accordance with specification MIL-F-14072 and the equipment drawings. (See 4.4)

3.5 Marking.

3.5.1 General. Marking shall conform to specification MIL-M-13231 (See 4.4) Front panel marking shall be group 1 as described in that specification.

3.5.2 Visibility. Wherever practicable, parts and components shall be so mounted that their identification markings will be readily visible with minimum disassembly. of the equipment.

3.5.3 Serial numbers. All items listed in 1.1 shall be separately serial numbered consecutively including transit cases.

3.6 Air-seal test, provision for. The transit case shall provide means for enabling performance of the air-seal test. (See 4.7)

3.7 Interchangeability. (See 4.8)

3.7.1 General. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.7.2 Filter interchangeability. Any Filter F-1482()/GRC shall be interchangeable with any other F-1482()/GRC in any input port in Coupler, CU-2266()/GRC (to form Multiplexer TD-1288()/GRC) or any input port in Coupler, CU-2267()/GRC (to form Multiplexer TD-1289()(V)/GRC).

3.7.3 Termination unit interchangeability. Any Termination Unit MX-10080()/GRC shall be interchangeable with any MX-10080()/GRC or Filter, F-1482()/GRC in any input port in Coupler, CU-2267()/GRC (to form Multiplexer TD-1289()(V)2/GRC or Multiplexer TD-1289()(V)3/GRC).

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3.8 Safety engineering.

3.8.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection. (See 4.4) A grounding stud for protection from lighting shall be a part of the unit design.

3.8.2 Radioactive materials. No radioactive material shall be utilized.

3.8.3 Edge rounding, exposed. Exposed edges and corners shall be rounded sufficiently to minimize laceration/puncture hazards, the design goal being a minimum radius of 0.04 inch (1 mm) for edges and 0.5 inch (12.7 mm) for corners.

3.9 (U) System nuclear survivability. All contractor-furnished mechanical configurations, electronic assemblages, electronic equipments, electronic circuits, and electronic pieceparts shall withstand the nuclear environments as specified in Appendix A (classified statement of criteria). If nuclear hardening fixes are deemed necessary to ensure the nuclear survivability of the equipments, the contractor shall implement the hardening fixes and shall demonstrate through the use of appropriate validation techniques that the equipments satisfy the specified nuclear criteria herein. The equipments shall meet the operational requirements of Section 3 herein, immediately after being subjected to the specified nuclear environments. This recovery in operational effectiveness may include automatic or manual correction of nuclear environment-induced changes in the system but not the replacement of pieceparts or components. If nuclear environmental testing is required to verify the nuclear survivability of the production equipments, the equipments shall be energized and the CFE and the GFE configured for the worst-case system response during exposure. The contractor shall ensure that the GFE is not degraded by nuclear environment-induced signals coupled from the CFE; and, vice versa, that the CFE is not degraded by nuclear environment-induced signals coupled from the GFE.

3.10 Electrical characteristics.

3.10.1 Multiplexer, specified performance - 50 ohms. The nominal load impedance for the multiplexer shall be 50 ohms real. The nominal output impedance shall be 50 ohms.

3.10.1.1 Insertion loss. Maximum multiplexer Insertion Loss shall be 1.75 dB for the TD-1288()/GRC and 2.00 dB for the TD-1289()(V)/GRC. (See 4.5.1.1.1)

3.10.1.2 Isolation. The multiplexers shall provide at least 40 dB isolation between F-1482()/GRC transceiver input ports when the F-1482()/GRC's are tuned to frequencies separated by 5% or more. (See 4.5.1.1.2)

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3.10.1.3 Passband. During operation, the passband of each multiplexer channel shall extend at least 100 kHz on each side (200 kHz minimum passband) of the tune frequency. Specified performance shall be maintained within this passband. (See 4.5.1.1.3)

3.10.1.4 Power handling capability. The multiplexer shall be capable of accepting a continuous input of 60 watts simultaneously into each input terminal and shall continue to operate within the limits in this specification. The associated transmitter output impedance shall be a nominal 50 ohms. (See 4.5.1.1.4)

3.10.1.5 Harmonics. Harmonics generated within the multiplexer shall be down more than 120 dB from a 60 watt fundamental signal. (See 4.5.1.1.5)

3.10.1.6 Intermodulation. Third order intermodulation products generated within the multiplexer in the range from 30 MHz to 50 MHz shall be attenuated no less than 115 dB below the desired signal levels while third order intermodulation products above 50 MHz up to 88 MHz shall be attenuated no less than 120 dB below the desired signal levels. (See 4.5.1.1.6)

3.10.1.7 Bandpass Filter, F-1482()/GRC, General Requirements - 50 Ohms.

3.10.1.7.1 Tuning range. Each bandpass filter when installed in a multiplexer configuration shall be capable of being manually tuned to any frequency within the 30 MHz to 88 MHz range. (See 4.5.1.1.7)

3.10.1.7.2 Coarse tuning. Each bandpass filter shall have a visual tuning indicator which will permit the bandpass filter to be coarse tuned to the desired operating frequency without the application of RF power. (See 4.5.1.1.7)

3.10.1.7.3 Fine tuning. Each bandpass filter shall have a meter and a meter function switch which will permit the bandpass filter to be fine tuned to the desired operating frequency directly from the coarse tuned preset position while applying RF power to the bandpass filter transceiver input port. (See 4.5.1.1.7)

3.10.1.7.4 Tuning power capacity. Each bandpass filter must be tunable using transmitters with nominal output powers ranging from 2 watts to 60 watts. (See 4.5.1.1.7)

3.10.1.7.5 Wattmeter calibration. The wattmeter on each bandpass filter shall be calibrated at full scale for both the 6 watt and 60 watt ranges. (See 4.5.1.1.8)

3.10.1.7.6 Tuning accuracy. When tuned to an operating frequency, the visual tuning indicator shall be within ±5 MHz of the desired frequency. (See 4.5.1.1.9)

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3.10.2 Multiplexer, specified performance - 3:1 VSWR. The nominal load impedance for any multiplexer shall be 50 ohms real. The specified performance shall be maintained when operating at a load VSWR of up to 3:1.

3.10.2.1 Power handling capability. The multiplexers shall be capable of accepting a continuous input of 60 watts simultaneously into each input terminal and shall continue to operate within the limits in this specification. The associated transmitter output impedance shall be a nominal 50 ohms. (See 4.5.2.1.1)

3.10.2.2 Bandpass Filter, F-1482()/GRC, general requirements - 3:1 VSWR.

3.10.2.2.1 Tuning range. Each bandpass filter when installed in a multiplexer configuration shall be capable of being manually tuned to any frequency within 30 MHz to 88 MHz range. (See 4.5.2.1.2)

3.10.2.2.2 Coarse tuning. Each bandpass filter shall have a visual tuning indicator which will permit the bandpass filter to be coarse tuned to the desired operating frequency without the application of RF power. (See 4.5.2.1.2)

3.10.2.2.3 Fine tuning. Each bandpass filter shall have a meter and a meter function switch which will permit the bandpass filter to be fine tuned to the desired operating frequency directly from the coarse tuned preset position while applying RF power to the bandpass filter transceiver input port. (See 4.5.2.1.2)

3.10.2.2.4 Tuning power capacity. Each bandpass filter must be tunable using transmitters with nominal output powers ranging from 2 watts to 60 watts. (See 4.5.2.1.2)

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3.11 Service conditions. The multiplexer shall meet the following service conditions; where a test is referenced, meeting the test shall be considered as compliance with the requirement. Where electrical operation of a multiplexer is required in conjunction with the conduct of a service condition test, electrical measurements shall be made on the multiplexer in accordance with Table I. Satisfactory compliance with the electrical measurements of Table I shall constitute conformance to the requirement in each applicable service condition test specified below for the multiplexer to "be operable without degradation in specified performance". In addition, the transit cases shall suffer no mechanical damage when used in conjunction with any multiplexer service condition test.

TABLE I

Service Condition	Elec Measurements before exposure (See 4.25.1)	Elec Measurements during exposure (See 4.25.2)	Elec Measurements after exposure (See 4.25.3)
High temperature	applicable	applicable*	applicable
Low temperature	applicable	applicable	applicable
Humidity	applicable	applicable	applicable
Altitude	applicable	applicable*	applicable
Rain	applicable	not applicable**	applicable
Dust	applicable	not applicable	applicable
Vibration (sine)	applicable	not applicable	applicable
Vibration (bounce, loose cargo)	applicable	not applicable	applicable
Shock (bench handling)	applicable	not applicable	applicable
Shock (drop)	applicable	not applicable	applicable
Shock (ballistic impact)	applicable	not applicable	applicable
Fungus	applicable	not applicable	applicable
Leakage	not applicable	not applicable	not applicable
Salt fog	not applicable	not applicable	not applicable

* Power handling capability test with a 50 ohm load to be performed, see 4.5.1.1.4.

** The electrical measurements of 4.25.2 are not to be performed on the multiplexer during the "Rain" test. However, the test of 4.5.1.1.4 shall be performed on the multiplexer during the "Rain" test.

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3.11.1 Temperature. (See 4.9)

3.11.1.1 Operating. The multiplexer shall be operable without degradation in specified performance at ambient temperatures in the range of +160°F (+125°F plus the effects of solar radiation at the rate of 360 BTU/ft^2/hr) and -50°F.

3.11.1.2 Non-operating. The multiplexer shall withstand exposure to ambient air temperature in the range of +160°F to -70°F.

3.11.2 Humidity. The multiplexer shall be operable without degradation in specified performance and shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials during the operating times specified in 4.10. Discoloration of the metal shall not be considered a failure.

3.11.3 Altitude (elevation). The multiplexer shall be operable without degradation in specified performance after it is tested as specified in 4.11.

3.11.4 Leakage (immersion). The transit case containing the multiplexer shall show no evidence of leakage after it is tested as specified 4.12.

3.11.5 Rain. The multiplexer shall be operable without degradation in specified performance after it is tested as specified in 4.13.

3.11.6 Dust. The multiplexer shall be operable without degradation in specified performance after it is tested as specified in 4.14.

3.11.7 Salt fog. After the salt-fog test of 4.15, the multiplexer when examined visually with the aid of a 10 power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surface; or exfoliation of metal (See 6.4). Electrical operation is not required. Discoloration of the metal shall not be considered a failure.

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3.11.8 Vibration, bounce, and shock. The multiplexer shall meet the requirements of Table II. Tests for vibration (bounce, loose cargo) and Shock (drop) shall be performed with the multiplexer in its transit case. Except for minor surface abrasion, the transit case shall exhibit no physical damage. The multiplexer shall not require retuning in order to be operable without degradation in specified performance after test.

TABLE II

Test Paragraph	Paragraph Heading	Performance after test
4.16	Vibration (sine)	Must be operable without degradation in specified performance and have sustained no physical or mechanical damage.
4.17	Vibration (bounce, loose cargo)	Must be operable without degradation in specified performance and have sustained no physical or mechanical damage.
4.18	Shock (bench handling)	Must be operable without degradation in specified performance and have sustained no physical or mechanical damage.
4.19	Shock (drop)	Must be operable without degradation in specified performance and have sustained no physical or physical damage.
4.20	Shock (ballistic impact)	Must be operable without degradation in specified performance and have sustained no physical or mechanical damage.

3.11.9 Fungus. After the fungus test of 4.21, the multiplexer shall be visually examined using a 10-power magnifier. The multiplexer shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth as evidenced by growth colonization which includes branching and sporulation (6.4) on or within each cubic foot, or fraction thereof equipment assembly volume. Isolated instances of partial tubular germination shall not be included in this evaluation. Areas where fungal growth has been supported by nutrients or contaminants as a result of handling or equipment exposure but fungal growth has not been sustained, are exempt from consideration. (See 6.4) Electrical operation is not required.

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3.12 Preconditioning. The equipment shall meet the requirements of 3.10 after being subjected to the preconditioning specified in 4.6.

3.13 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs. (See 4.22)

a. In MIL-P-11268:

General requirements for plastic material and plastic parts.	Cleaning
Wiring and cabling, including:	General requirements for securing of parts.
Slack	+Process for welding.
Protection	Cleaning after welding
Clearance	Controls
Splicing	Self-tapping screws.
Connections, general	
Grounding, general	+Cleaning prior to welding.
Shielding on wire and cable	
Soldering and brazing.	

b. In this specification:

Construction (3.2)

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirement specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Any test equipment required for government inspection shall be made available by the contractor.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include packaging.

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging, (See 4.5)

(2) Quality conformance inspection of packaging. (See 4.23)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish 6 first article units of the complete transceiver multiplexer TD-1289()(V)/GRC and 6 first article units of the complete transceiver multiplexer TD-1288()/GRC.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table III, and shall be performed in the order specified therein.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE III

FIRST ARTICLE INSPECTION

Inspections	Reqt Para	Test Para	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6
1. Inspections covered by subsidiary documents	3.3,-3.5, 3.8	4.4	Inspections to be performed on all units.					
2. Preconditioning	3.12	4.6	Inspections to be performed on all units.					
3. Group A inspection	(See Table IV)		Inspections to be performed on all units.					
4. Group B inspection	(See Table V)		Inspections to be performed on all units.					
5. Nuclear Survivability	3.9	4.24	1					
6. Environmental								
Low temperature	3.11.1	4.9.1		1				
High temperature	3.11.1	4.9.2		2				
Humidity	3.11.2	4.10		3				
Altitude	3.11.3	4.11			1			
Rain	3.11.5	4.13	2					
Dust	3.11.6	4.14	3					
Salt Fog	3.11.7	4.15				1		
Vibration	3.11.8							
Sine		4.16			2			
bounce, loose cargo		4.17			3			
Shock	3.11.8							
bench handling		4.18						1
drop		4.19						2
ballastic impact		4.20						3
Fungus	3.11.9	4.21					1	

Notes:

a. Unit as delineated above shall be construed as 1 each TD-1288()/GRC and 1 each TD-1289()(V)1/GRC. The latter may be modified at government discretion to form the TD-1289()(V) 2/GRC or the TD-1289()(V)3/GRC.

b. The inspections 1 through 4, in the order shown shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

c. The implementation of inspections covered in 5 & 6 shall be expedited to satisfy contract delivery requirements.

d. The numbers in the unit columns in the table covered by 5 & 6 specify the order of inspections for the indicated unit.

e. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after the salt fog and fungus tests before proceeding with subsequent inspection.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Parts, Materials and Processes:	
General	3.3
Finish	3.4
Marking	3.5
Safety	3.8

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.4, and 4.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table IV. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures and the general inspection levels of MIL-STD-105, and the AQL's indicated in Table IV.

4.5.1.1 Electrical measurements.

4.5.1.1.1 Insertion Loss. Tune the bandpass filter for maximum return loss (corresponds to minimum VSWR/reflected power) by use of a display scope and a sweep RF signal generator containing a marker signal. Using the measured return loss (Lr), calculate the VSWR.

$$VSWR = \frac{10^{Lr/20} + 1}{10^{Lr/20} - 1}, \quad Lr = 10 \log \frac{P \text{ forward}}{P \text{ reflected}}$$

Determine the insertion loss by noting the difference in signal levels displayed on the scope both when the multiplexer is inserted in the signal path and when the multiplexer is removed from the signal path.

The bandpass filter shall then be retuned to the next marker signal frequency and its insertion loss also determined. Repeat the preceeding procedures for the remaining marker signal frequencies. Repeat the preceeding for each bandpass filter in the multiplexer. The marker signal frequencies are 30.0MHz, 44.5MHz, 59.0MHz, 63.4MHz and 88.0MHz.

4.5.1.1.2 Isolation. Tune each pair of bandpass filters to the specified test frequencies. This is done by tuning the bandpass filter for maximum return loss (corresponds to minimum VSWR/reflected power) by use of a display scope and a sweep RF signal generator containing a marker signal. Using the measured return loss (Lr) calculate the VSWR. Insert a signal into the bandpass filter tuned to the higher frequency and detect the signal with an RF voltmeter connected to the bandpass filter tuned to the lower frequency. Assuming that the available bandpass filters are numbered from left to right, the test frequencies are:

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Bandpass Filter Numbers	Resonant Frequencies in MHz
1 & 2	30.4/32.0
1 & 2	49.4/52.0
1 & 2	72.2/76.0
1 & 2	83.6/88.0
1 & 3	30.4/32.0
1 & 3	49.4/52.0
1 & 3	72.2/76.0
1 & 3	83.6/88.0
3 & 4	30.4/32.0
3 & 4	49.4/52.0
3 & 4	72.2/76.0
3 & 4	83.6/88.0
1 & 5	30.4/32.0
2 & 5	49.4/52.0
3 & 5	72.2/76.0
4 & 5	83.6/88.0

4.5.1.1.3 Passband. Tune the bandpass filter for maximum return loss (corresponds to minimum VSWR/reflected power) by use of a display scope and a sweep RF signal generator containing a marker signal. Using the measured return loss (Lr), calculate the VSWR.

Without further adjustment or tuning to the bandpass filter, vary the marker signal frequency both above and below the marker signal frequency used to tune the bandpass filter and note the frequencies where the insertion loss fails to meet specified requirements. The bandpass filter shall then be retuned to the next marker signal frequency and the above procedure is repeated. The frequencies where the insertion loss fails to meet the specified requirements are noted. Repeat the procedure for the remaining marker frequencies. Repeat the preceeding for each bandpass filter in the multiplexer. The marker signal frequencies are 30.0MHz, 44.5MHz, 59.0MHz, 73.5MHz and 88.0MHz.

4.5.1.1.4 Power handling capability. Tune each bandpass filter to its specified test frequency by means of its built-in tuning indicator. Operate all bandpass filters simultaneously at 60 watts indicated forward power as measured by an external thru-line wattmeter on the transceiver input port. The duration of simultaneous operation shall be 10 minutes. Depending on the number of bandpass filters in the multiplexer, the order of usage of test frequencies are 30.0MHz, 88.0MHz, 44.5MHz, 73.5MHz and 59.0MHz.

4.5.1.1.5 Harmonics. Tune the bandpass filter by means of its self-contained tuning indicator. Operate the bandpass filter with 60 watts indicated forward power as measured by a thru-line watt meter on the transceiver input port. Using a notch filter to minimize the fundamental signal, determine the 2nd, 3rd, 4th and 5th harmonics as indicated by spectrum analyzer displays of the load plus the multiplexer and the load alone (with multiplexer out of the circuit). The tune frequencies for the bandpass filter are 30.0MHz, 44.5MHz, 59.0MHz, 73.5MHz and 88.0MHz. Repeat the preceeding for each bandpass filter in the multiplexer.

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4.5.1.1.6 Intermodulation. Tune a bandpass filter for maximum return loss (corresponds to minimum VSWR/reflected power) by use of a display scope and a sweep RF signal generator containing a marker signal. Using the measured return loss (Lr), calculate the VSWR. Use a second bandpass filter for maximum return loss as above and calculate its VSWR.

Operate both bandpass filters simultaneously at 60 watts indicated forward power as measured by external thru-line watt meter on the transceiver input ports. Measure and note the upper and lower product levels as displayed on a spectrum analyzer. Connect the signal generator, set in turn to both product frequencies, to the spectrum analyzer and adjust the signal generator output level to give the same display on the spectrum analyzer as the product level. Record the microvolt level from the signal generator and convert to dB. The test frequencies (in MHz) are:

Bandpass Filter #1	Bandpass Filter #2	Lower Product	Upper Product
32.3	34.0	30.6	35.7
49.4	52.0	46.8	54.6
72.2	76.0	68.4	79.8
78.85	83.0	74.7	87.15

Repeat using Bandpass Filters #1 and #3
Repeat using Bandpass Filters #3 and #4
Repeat using Bandpass Filters #1 and #5

4.5.1.1.7 Tuning power capacity. With 2 watts indicated forward power applied, as measured by an external thru-line watt meter on the transceiver input ports tune each bandpass filter to the specified frequencies by means of its built-in tuning indicator. The specified frequencies are 30.0MHz, 44.5MHz, 59.0MHz, 73.5MHz and 88.0MHz. Repeat for each bandpass filter in the multiplexer. Repeat for 60 watts forward power applied, as measured by an external thru-line watt meter on the transceiver input ports.

Satisfactory completion of this test will also verify specification conformance to the requirements for "Tuning Range", para 3.10.1.7.1; "Coarse tuning", para 3.10.1.7.2; and "Fine tuning", para 3.10.1.7.3.

4.5.1.1.8 Wattmeter calibration. Tune the bandpass filter to the specified frequencies by means of its built-in tuning indicator. With specified power applied, as measured by an external thru-line wattmeter on the transceiver input port, record the power levels indicated on the bandpass filters built-in meter. The specified frequencies are 30MHz, 44.5MHz, 59.0MHz, 73.5MHz and 88.0MHz and the specified powers are 6 watts with full scale meter of 6 watts and 60 watts with full scale meter of 60 watts. Repeat for each bandpass filter in the multiplexer.

4.5.1.1.9 Tuning accuracy. Tune the bandpass filter to the specified frequencies by means of its built-in tuning indicator. Record the center frequencies indicated in each bandpass filters upper, middle and lower visual tuning window. The specified frequencies are 30MHz, 44.5MHz, 59.0MHz, 73.5MHz and 88.0MHz. Repeat for each bandpass filter in the multiplexer.

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4.5.1.2 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

TABLE IV
GROUP A INSPECTION

Inspection	Requirement Paragraph	Test Paragraph	AQL Major	AQL Minor
Visual and Mechanical	3.2, 3.10.2.2, 3.10.2.3, & 3.13	4.22	10 dphu*	15 dphu*
Electrical (50Ω)			1%	**
Insertion loss	3.10.1.1	4.5.1.1.1		
Isolation	3.10.1.2	4.5.1.1.2		
Passband	3.10.1.3	4.5.1.1.3		
Power Handling Capibility	3.10.1.4	4.5.1.1.4		
Harmonics	3.10.1.5	4.5.1.1.5		
Intermodulation	3.10.1.6	4.5.1.1.6		
Tuning Power Capacity	3.10.1.7.4	4.5.1.1.7		
Tuning Range	3.10.1.7.1	4.5.1.1.7		
Coarse tuning	3.10.1.7.2	4.5.1.1.7		
Fine tuning	3.10.1.7.3	4.5.1.1.7		
Wattmeter Calibration	3.10.1.7.5	4.5.1.1.8		
Tuning Accuracy	3.10.1.7.6	4.5.1.1.9		
Air Seal	3.6	4.7	1%	

* Defects per hundred units
** Defects in all characteristics other than visual and mechanical are considered major.

4.5.2 Group B Inspection. Group B inspection shall be performed in inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table V and to the special inspection levels of Table I to MIL-STD-105.

4.5.2.1 Electrical measurements

4.5.2.1.1 Power handling capability. This test shall be performed using a load which produces a VSWR of 3:1 in an otherwise 50 ohm system. Tune each bandpass filter to its specified test frequency by tuning for maximum return loss (corresponds to minimum VSWR/reflected power) by use of a display scope and a sweep RF signal generator containing a marker signal. Using the

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measured return loss (Lr), calculate the VSWR. Operate all channels simultaneously for 10 minutes at 60 watts indicated forward RF power as measured by an external thru-line wattmeter connected to the transceiver input. Depending on the number of bandpass filters in the multiplexer, the order of usage of test frequencies is 30.0MHz, 88.0MHz, 44.5MHz, 73.5MHz and 59.0MHz.

4.5.2.1.2 Tuning power capacity. This test shall be performed in accordance with paragraph 4.5.1.1.7 of this specification except that the load impedance shall produce a VSWR of 3:1. Satisfactory completion of this test will also verify specification conformance to the requirements for "Tuning Range", para 3.10.2.2.1; "Coarse Tuning", para 3.10.2.2.2; and "Fine Tuning", para 3.10.2.2.3.

4.5.2.2 Group B sampling plan. The Group B AQL shall be 4 percent defective and the inspection level shall be S-4.

4.5.2.3 Order of inspection within Group B. Group B Inspection shall be performed in an order satisfactory to the Government.

TABLE V
GROUP B INSPECTION

Inspection	Requirement Paragraph	Test Paragraph	AQL
Electrical (3:1 VSWR)			*
Power Handling Capability	3.10.2.1	4.5.2.1.1	4%
Tuning Power Capacity	3.10.2.2.4	4.5.2.1.2	4%
Tuning range	3.10.2.2.1	4.5.2.1.2	4%
Coarse tuning	3.10.2.2.2	4.5.2.1.2	4%
Fine tuning	3.10.2.2.3	4.5.2.1.2	4%
Interchangeability	3.7	4.8	4%
Leakage	3.11.4	4.12	4%

* All electrical defects are major.

4.5.3 Group C inspection. Group C Inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table VI. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1.1 Six (6) samples of the complete equipment (see Note a, Table III) shall be selected at random, one(1) sample for each subgroup 1 through subgroup 6 in Table V. The samples shall be selected at the start of the contract from the first quality conformance inspection lot. These samples shall constitute the Group C requirement for the first 100 units produced.

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4.5.3.1.2 Thereafter, 4 samples of the complete equipment (see Note a, Table III), shall be selected at random, one (1) sample for each subgroup 1 through subgroup 4 in Table VI from every 200 units and fraction thereof produced.

4.5.3.2 Order of inspection Within Group C. Group C Inspection shall be performed in accordance with the sequence given in Table III for the environmental tests.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

TABLE VI
GROUP C INSPECTION

Inspection	Requirement Paragraph	Inspection Paragraph
Subgroup 1		
Rain	3.11.5	4.13
Dust	3.11.6	4.14
Subgroup 2		
Low temperature	3.11.1	4.9.1
High temperature	3.11.1	4.9.2
Humidity	3.11.2	4.10
Subgroup 3		
Altitude	3.11.3	4.11
Vibration	3.11.8	
Sine		4.16
Bounce, loose cargo		4.17
Subgroup 4		
Shock	3.11.8	
bench handling		4.18
drop		4.19
ballistic impact		4.20
Subgroup 5		
Salt Fog *	3.11.7	4.15
Subgroup 6		
Fungus *	3.11.9	4.21

*The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

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4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A and Group B inspections, in that order.

4.6 Preconditioning. Prior to Groups A, B, and C inspection, each equipment on order shall be subjected to a 24 hour preconditioning period after which the equipment shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to subsequent inspection. Prior to preconditioning, each equipment shall meet full specification performance for the measurements specified in 3.10.

4.6.1 Test level. The preconditioning shall be conducted under the environmental conditions specified for "ground vehicle" equipments in Table I, MIL-STD-781C.

4.6.2 Equipment duty cycle. During the preconditioning period, the unit shall not be operated.

4.6.3 Environmental profile. The environmental profile of the preconditioning period shall be as shown in Figure I, MIL-STD-781C.

4.7 Air-seal test. The transit case shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the transit case shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the transit case) applied to the interior of the transit case. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period, the decrease in vacuum shall not exceed 0.01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily.

4.8 Inspection for dimensional interchangeability.

Interchangeability dimensional criteria as listed below shall be gaged or measured to determine conformance with the requirements of 3.7. Government gages shall be used if they are furnished on the contract. When a listed dimension is not within specified tolerance or design limit, it shall be considered a major defect.

a. Maximum overall dimensions of Bandpass filter modules and/or termination unit modules.

b. Minimum cavity areas for the modules on the mountings.

c. Location of connectors on modules and mounting cavities.

d. Location of mounting holes and fasteners, on modules, and cavities.

e. Location of bayonet pins or holes (if provided) on modules and cavities.

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4.9 Temperature test.

4.9.1 Low temperature. The equipment shall be tested in accordance with Method 502.1 Procedure I, of MIL-STD-810. The storage temperature shall be -70°F, and the operating temperature shall be -50°F. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.10.

4.9.2 High temperature. The equipment shall be tested in accordance with Medthod 501.1 Procedure II, of MIL-STD-810. The operating temperature shall be 160°F. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.10.

4.10 Humidity. The multiplexer shall be subjected to the humidity test specified in Method 507.1 Procedure II, of MIL-STD-810, with test measurements during Step 4 and the last 5-hour period of the fifth 48-hour cycle in Step 6. Inability of the multiplexer to meet the requirements of 3.11.2 shall constitute failure of this test. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.10.

4.11 Altitude (elevation) test. The equipment shall be tested in accordance with Method 500.1 Procedure I, of MIL-STD-810. The equipment shall be tested to withstand an altitude of 40,000 feet, non-operating. Prior to testing, the equipment shall meet full specification performance for the measurements in 3.10.

4.12 Leakage (immersion) test. The transit case containing the multiplexer shall be tested in accordance with Method 512.1 Procedure I, of MIL-STD-810.

4.13 Rain test. The equipment shall be tested as follows:

4.13.1 Dry at 150° ± 5°F for 48 hours.

4.13.2 Condition for 4 hours at 77° ±5°F and 40 to 60 percent relative humidity.

4.13.3 Take pretest data. The equipment shall meet the electrical requirements specified in Table I.

4.13.4 Test equipment in accordance with Method 506.1 Procedure I, of MIL-STD-810. The equipment shall be operated during the last 10 minutes of each 30 minute test period. Complete final measurements within 1 hour after the rain has been discontinued.

4.13.5 Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.10.

4.14 Dust test. The equipment shall be tested in accordance with Method 510.1 Procedure I, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.10.

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4.15 Salt fog. The multiplexer shall be subjected to the salt fog test specified in Method 509.1 Procedure I, of MIL-STD-810. No corrosion products shall be deposited on the test item by the facility. Inability of the multiplexer to meet the requirements of 3.11.7 shall constitute failure of this test.

4.16 Vibration (sine). The equipment shall be tested in accordance with Method 514.2 Procedure VIII, of MIL-STD-810. Test levels and conditions shall be as specified in Table 514.2-VI for tracked vehicles. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.10.

4.17 Vibration (bounce, loose cargo). The equipment shall be tested in accordance with Method 514.2 Procedure XI, Part 2 of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.10.

4.18 Shock test; bench handling. The equipment shall be tested in accordance with Method 516.2 Procedure V, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.10.

4.19 Shock test; drop. The equipment shall be tested in accordance with Method 516 2 Procedure II, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.10.

4.20 Shock; ballistic impact. The equipment shall be subjected to the test of MIL-S-901 for type A, lightweight equipment. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.10.

4.21 Fungus. The multiplexer shall be subjected to the fungus test specified in Method 508.1 Procedure I, of MIL-STD-810. There shall be no abundant growth colonization on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the multiplexer is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the multiplexer to meet the requirements of 3.11.9 shall constitute failure of this test.

4.22 Visual and mechanical inspection. Each multiplexer including accessories, if any, and transit case shall be examined for the defects listed in standard MIL-STD-252 and those features specified in paragraphs 3.2 and 3.13. Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of MIL-STD-105.

4.23 Quality conformance inspection of packaging. The packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirement of Section 5.

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4.24 (U) Nuclear hardness validation.

4.24.1 (U) Test and Analyses. Tests and analyses may be necessary on the equipment in its intended operating configuration to assure the system nuclear survivability as specified in paragraph 3.9. Specific areas of consideration are discussed in paragraphs 4.24.2 thru 4.24.5 inclusive.

Calibration certification of the test facilities to be used are required by the government, 45 days prior to nuclear test. Verification of the wave shape of the test simulator is required. Calibration and wave shape verification shall be furnished to the government by the contractor.

4.24.2 (U) Electromagnetic pulse. The equipment shall be subjected to the EMP environments as specified in Appendix A. The equipment shall be placed within a ground-effect electromagnetic pulse facility in the following orientations:

a. Connecting cables extended parallel to the electric field and perpendicular to the magnetic field; all individual units must be in the test volume but widely separated.

The equipment shall be exposed at threat level in the above orientations and in each operating mode of the equipment. The equipment shall meet the operational requirements of Section 3 upon completion of threat level tests. Where necessary, test data to satisfy the EMP requirements may be obtained by extrapolation of diagnostic test levels to threat level and injection pulsing at the calculated level.

4.24.3 (U) Initial nuclear radiation. For the purpose of the design, but not validation testing, electronic pieceparts and circuits for which nuclear test data are available or for which an acceptable level of confidence has been established through mathematical analysis need not be subjected to further nuclear design testing. Acceptance of these electronic pieceparts, circuit and mathematical analysis will be determined by the procuring agency from data supplied to the Government by the contractor.

4.24.3.1 Neutron fluence. The equipment shall be exposed in such a manner that all electronic pieceparts and circuits receive the Neutron Fluence specified in Appendix A. The equipment shall then perform as specified in Section 3.

4.24.3.2 Gamma total dose. The equipment shall be exposed in such a manner that all electronic pieceparts and circuits receive the Gamma Total Dose specified in Appendix A. The equipment shall be energized during exposure and shall then perform as specified in Section 3. Except for fiber optics and n-channel metal-oxide semiconductors (NMOS), verification of device hardness against permanent damage from gamma ray dose may be achieved by the gamma output which normally accompanies the neutron fluence. Gamma dose hardness of fiber optics and NMOS devices must be verified by Cobalt-60 irradiation in which the total dose specified is delivered in a steady-state exposure on the order of 20 seconds duration.

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4.24.3.3 Peak Gamma Dose Rate. The equipment shall be exposed in such a manner that all electronic pieceparts and circuits receive the Peak Gamma Dose Rate specified in Appendix A . The equipment shall be energized during exposure and shall then perform as specified in Section 3.

4.24.4 (U) Thermal radiation. One complete equipment will be exposed to thermal radiation levels as specified in Appendix A. If the utilized simulation facilities provide these thermal radiation levels only over relatively small areas, many individual exposures will be required so that each of the exposed surfaces (case, meters, knobs, antennas, cables, and any other outboarded components) will receive the required thermal exposure. Analysis of the equipment following exposure shall show no degradation that will affect the operation of the equipment equipment as required in Section 3. Sample sections may be used for this test.

4.24.5 (U) Nuclear-Air Blast. The equipment in its intended operating configuration will be subjected to Nuclear-Air testing. If the equipment is unprotected, then it shall meet the Air Blast environments as specified in Appendix A . If the equipment is housed within an enclosure, then the equipment shall be subjected to the induced shock which will be transmitted through the enclosure when the enclosure is subjected to the Air Blast environments as specified in Appendix A. In both cases the equipment shall meet the operational requirements of Section 3 herein following this test.

Tests or analysis shall be performed to account for synergistic effects of thermal radiation received before the nuclear-air blast.

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4.25 Service Condition test electrical measurements.

4.25.1 Electrical measurements before equipment exposure to an environmental service condition.

4.25.1.1 Tune each bandpass filter for maximum return loss (corresponds to minimum VSWR/reflected power) by use of a display scope and a sweep RF signal generator with a marker signal. Using the measured return loss (Lr), calculate the VSWR

$$VSWR = \frac{10^{Lr/20}+1}{10^{Lr/20}-1}, \quad LR = 10 \log \frac{P \text{ forward}}{P \text{ reflected}}$$

Depending on the number of bandpass filters in the multiplexer, the order of usage of test frequencies is 30.0MHz, 88.0MHz, 44.5MHz, 73.5MHz and 59.0MHz.

4.25.1.2 The reference level is the fixed signal generator output level determined by an RF voltmeter connected directly to the signal generator.

4.25.1.3 Determine the pre-test resonant frequency (F_o) of each filter in the multiplexer. This is accomplished by varying the signal generator frequency while maintaining a constant signal generator output level and observing the level at the multiplexer antenna port. The frequency (labeled Fa) above the bandpass filter tune frequency set in paragraph 4.25.1.1 and the frequency (labeled Fb) below the bandpass filter tune frequency set in paragraph 4.25.1.1 at which an RF voltmeter, connected to monitor multiplexer output signal level, indicates a signal level 10dB below the reference level determined in paragraph 4.25.1.2 are noted. The resonant frequency (F_o) is then calculated using the formula $F_o = \frac{Fa + Fb}{2}$

4.25.1.4 The signal generator is set to the calculated pre-test resonant frequency F_o and without any adjustment to the bandpass filter, each bandpass filter's insertion loss is determined by use of an RF voltmeter.

4.25.1.5 Repeat paragraph 4.25.1.4 for the signal generator set to F_o + 100KHz and F_o -100KHz.

4.25.2 Electrical measurements during equipment exposure to an environmental service condition.

4.25.2.1 Determine each bandpass filters resonant frequency (labeled Fe) during environmental exposure.

4.25.2.2 Set the signal generator to the pre-test resonant frequency (F_o) and determine each bandpass filters insertion loss.

4.25.2.3 Repeat paragraph 4.25.2.2 for the signal generator set to F_o +100KHz and F_o -100KHz.

4.25.3 Electrical measurements after equipment exposure to an environmental service condition. Same as electrical measurements (4.25.2.1 - 4.25.2.3) during equipment exposure to an environmental service condition.

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. JR-1-1-ØØ-00 NS-X-ED-Ø-00-A.

5.1.2 Level B. JR-1-1-ØØ-00 NS-X-ED-Ø-00-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55565.

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6. NOTES

6.1 Intended use. The Multiplexer TD-1288()/GRC allows two transceivers to be connected to a single broadband antenna. The three versions of the Multiplexer TD-1289()(V)/GRC allow from three to five transceivers to be connected to a single antenna. When either three or four transceivers are required to be connected to a single broadband antenna, then a Termination Unit MX-10080()/GRC may be substituted for the unused Bandpass Filter F-1482()/GRC module.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, Level B preservation and packing (See Section 5).

c. When rough handling test is required (See Section 4).

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature required, test tools and running spares.

g. First Article Inspection. (See 3.1)

h. Length of cables (if any are required).

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Definitions.

6.4.1 Branching. Branching is a connected arrangement of filaments (Hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.4.2 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.4.3 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes an abundant growth.

6.4.4 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

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6.4.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies called "spores".

6.4.6 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular-shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.5 Group C inspection. (See 4.5.3.3) Approval to ship may be withheld at the discretion of the government pending the decision from the contracting officer on the adequacy of corrective action.

6.5.1 Initial shipments. (See 4.5.3.3) Initial shipments of end items will be contingent only upon successful completion of all Group C tests.

6.6 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TD-1288A /GRC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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PREPARING ACTIVITY

ARMY-CR
PROJECT 5805-A290

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-49247(CR) Transceiver Multiplexers

TD-1289()/GRC and TD-1289() (V)/GRC

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

CDR, CORADCOM
ATTN: DRDCO-COM
Ft. Monmouth, N.J. 07703

☐ VENDOR ☐ USER ☐ MANUFACTURER

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